

We saw that you can multiply a vector and a scalar, but is there a useful way to "multiply" two vectors?

Dot Product (Inner Product)

Let $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle$

The dot Product of $\vec{a}$ and $\vec{b}$ is

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

as usual, this works for any dimension, say an n dimensional vector

$$\vec{u} \cdot \vec{v} = \sum_{i=1}^n u_i v_i$$

Ex:

$$\vec{a} = \langle 1, 2, 3 \rangle, \quad \vec{b} = \langle 4, 5, 6 \rangle$$

$$\begin{aligned}\vec{a} \cdot \vec{b} &= 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 \\ &= 4 + 10 + 16\end{aligned}$$

$$= 30$$

Properties of dot Product

Let $\vec{a}, \vec{b}, \vec{c}$ be vectors, $c \in \mathbb{R}$

$$1) \vec{a} \cdot \vec{a} = |\vec{a}|^2$$

$$2) \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

$$3) \vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

$$4) (c\vec{a}) \cdot \vec{b} = c(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (c\vec{b})$$

$$5) \vec{0} \cdot \vec{a} = 0$$

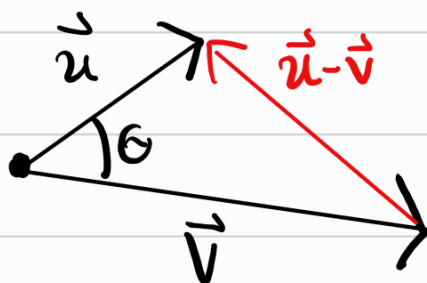
Dot Product have many uses, the following is the most used one:

Thm:

Let θ be the angle between vectors $\vec{u}$ and $\vec{v}$, then

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos(\theta)$$

Pf/ Consider the following picture for reference



From the law of cosine

$$|\vec{u} - \vec{v}|^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2|\vec{u}| |\vec{v}| \cos \theta$$

From earlier,

$$\begin{aligned} |\vec{u} - \vec{v}|^2 &= (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) = (\vec{u} - \vec{v}) \cdot \vec{u} - (\vec{u} - \vec{v}) \cdot \vec{v} \\ &= \vec{u} \cdot \vec{u} - \vec{v} \cdot \vec{u} - \vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{v} = |\vec{u}|^2 - 2\vec{u} \cdot \vec{v} + |\vec{v}|^2 \end{aligned}$$

So

$$|\vec{u}|^2 + |\vec{v}|^2 - 2\vec{u} \cdot \vec{v} = |\vec{u}|^2 + |\vec{v}|^2 - 2|\vec{u}||\vec{v}|\cos\theta$$

$$-2\vec{u} \cdot \vec{v} = -2|\vec{u}||\vec{v}|\cos\theta$$

$$\underline{\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}|\cos\theta}$$

Ex:

Say vectors $\vec{u}$ and $\vec{v}$ intersect at
an angle of $\frac{\pi}{3}$ and $|\vec{u}|=4, |\vec{v}|=6$
 $\cos(\frac{\pi}{3})$

$$\vec{u} \cdot \vec{v} = 4 \cdot 6 \cdot \frac{1}{2} = 12$$

The main way we use dot product is to
find the angle θ as

$$\cos(\theta) = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}||\vec{v}|}$$

Note: Two vectors are orthogonal if and only if
their dot product is 0.

Ex:

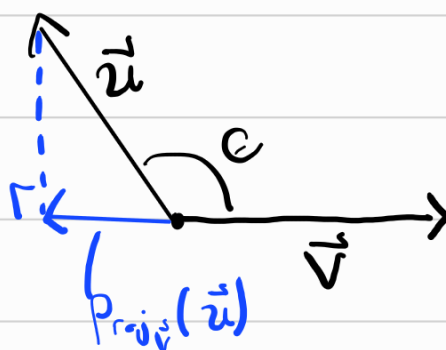
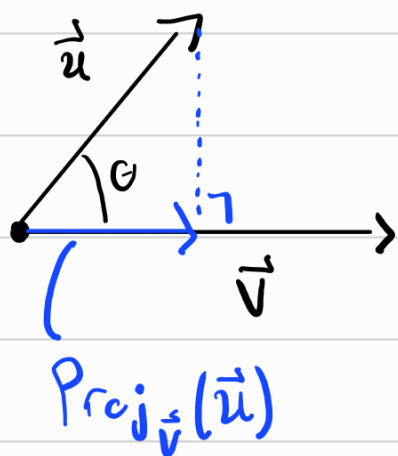
$$\vec{u} = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}, \quad \vec{v} = 5\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$$

$$\begin{aligned}\vec{u} \cdot \vec{v} &= 2 \cdot 5 + 2 \cdot (-4) + (-1) \cdot 2 \\ &= 10 - 8 - 2 \\ &= 0\end{aligned}$$

So $\vec{u}$ and $\vec{v}$ are orthogonal

Projections:

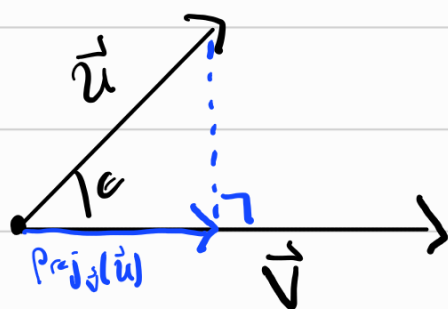
Take the vectors $\vec{u}$ and $\vec{v}$, we can look at the shadow of one vector on the other



$\text{Proj}_{\vec{v}}(\vec{u})$ is the Vector Projection of $\vec{u}$ onto $\vec{v}$

The signed magnitude of the vector projection is called a Scalar Projection denoted $\text{Comp}_{\vec{v}}(\vec{u})$

Drawing a triangle



$$\cos(\theta) = \frac{|\text{Proj}_{\vec{v}}(\vec{u})|}{|\vec{u}|}$$

$$\frac{\vec{u} \cdot \vec{v}}{|\vec{u}||\vec{v}|} = \frac{\text{Comp}_{\vec{v}}(\vec{u})}{|\vec{u}|}$$

$$\text{Comp}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|}$$

$\rightarrow$

$$\text{Proj}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \cdot \vec{v}$$

Ex:

$$\vec{u} = \langle 1, 1, 2 \rangle, \quad \vec{v} = \langle -2, 3, 1 \rangle$$

$$|\vec{u}| = 2, \quad |\vec{v}| = \sqrt{14}$$

$$\vec{u} \cdot \vec{v} = -2 + 3 + 2 = 3$$

$$\text{Comp}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} = \frac{3}{\sqrt{14}}$$

$$\text{Proj}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \cdot \vec{v} = \frac{3}{\sqrt{14}^2} \cdot \vec{v} = \frac{3}{14} \vec{v}$$

$$= \left\langle \frac{-6}{14}, \frac{9}{14}, \frac{3}{14} \right\rangle$$

$$= \left\langle -\frac{3}{7}, \frac{9}{14}, \frac{3}{14} \right\rangle$$